Linearity Indices and Linearity Improvement of 2-D Tetralateral Position-Sensitive Detector

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***Abstract*—A 2-D tetralateral position-sensitive detector (PSD) is able to provide continuous position measurements with high position resolution, fast response, and low cost. However, it suffers from the nonlinearity problem, which leads to inaccurate measurements. In this paper, two linearity indices are proposed as measures of linearity for its position measurement. Then, a systematic procedure is developed to design new formulas to achieve any degree of linearity through the use of the linearity indices. As an example, a simple estimation formula for the 2-D tetralateral PSD has been derived based on the procedure. Compared to the conventional formula, the linearity of the measurement is greatly improved when the new formula is used. To verify its effectiveness, the new formula is implemented in analog circuits and tested in our experiment. Similar linearity improvement for other 2-D PSDs can also be achieved by following the same design procedure.**

***Index Terms*—Linearity improvement, optical position measurement, position-sensitive detector (PSD), signal processing.**

I. Introduction

A 2-D tetralateral position-sensitive detector (PSD) is capable of providing continuous position measurement of the incident light spot with high sensitivity and resolution in two dimensions. It consists of a single square p-n junction with a resistive layer. Four electrodes are placed along each side of the square near its boundary on the same surface. When the sensor is illuminated nonuniformly, a position-dependent potential is generated [1] due to the lateral photo effect [2], which results in current flow on the resistive layer under the fully reverse-biased condition [3]. Photocurrents can be collected and measured at each electrode to estimate the position of the incident light in two dimensions via an intuitive formula. Other than the 2-D tetralateral PSD, a 2-D duolateral PSD, where two electrodes opposite each other are placed on the p-surface and the other two are on the n-surface of a square p-n junction [4], and a four-quadrant (4Q) PSD, which consists of four photodiodes positioned symmetrically around the center and separated by small gaps [5], can also provide continuous 2-D position measurement of the incident beam. This group of 2-D sensors has wide applications in position and angle sensing, vibration measurement, optical range finder, camera autofocusing, autonomous navigation, and atomic force microscopy (AFM) [6]–[8].

Compared to the 2-D duolateral PSD, the 2-D tetralateral PSD has the advantages of fast response, much lower dark current, easy bias application, and lower fabrication cost [9]. In addition, its measurement accuracy and resolution are independent of the spot shape and size, unlike the 4Q PSD that could easily be changed by air turbulence [5]. However, it suffers from the nonlinearity problem. Theoretically, the relationship between the position estimate and the real position is implicitly nonlinear. While the position estimate is approximately linear with respect to the real position when the spot is in the center area of the PSD, the relationship becomes nonlinear when the light spot is away from the center, and this results in severe estimation inaccuracy. This seriously limits its applications, and there are urgent demands for linearity improvement in many applications. For example, as a measurement tool of atomic interaction in the nanometer scale, AFM prefers the 4Q PSD when measuring the enlarged version of modulated atomic interaction between the probe and atoms because of its slightly better resolution than the 2-D tetralateral PSD. However, the measurement will only be accurate when the light spot is in the center area of the PSD due to the same nonlinearity problem in the 4Q PSD. Thus, the nonlinearity problem is a bottleneck to further resolution enhancement of AFM [10], and a PSD with a larger linear range and fast response is desired to further enlarge the atomic dynamics. The unknown spot shape and light distribution in the 4Q PSD make the linearity improvement extremely difficult, and hence, the 2-D tetralateral PSD is a potential candidate to achieve a breakthrough in linearity improvement. Attempts have been made to modify the contact configuration and to redesign the boundary shape of the 2-D tetralateral PSD for linearity improvement. Pin-cushion PSD with curvature in boundary [11] and clover PSD with more sophisticated design in boundary and contacts [12] are two representative technologies. Although the linearity of modified PSDs has been improved, their complicated structures make the manufacturing process complicated and result in a much higher cost. Moreover, it must be noted that the problem of surface inhomogeneities will further introduce unknown nonlinearities [13]. Other than changing the PSD structure, one may apply advanced signal processing techniques to the original 2-D tetralateral PSD when estimating the spot position. This approach will remove the concern of additional cost in manufacturing and other problems that a complicated PSD structure will bring. In the literature, estimation methods based on neutral network and interpolation algorithms have been developed [14], [15].

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However, in order to get an accurate estimate, they need a huge number of training data from the calibration process before the deployment of the PSD, which is time consuming. Furthermore, their tedious procedures and slow processing speed make them unrealistic in real implementations that require fast response like AFM. Thus, in order to achieve a better linearity performance without tradeoffs in cost and other factors, a new, simple, and implementable position formula is needed for the 2-D tetralateral PSD.

In this paper, after a brief review of the 2-D tetralateral PSD mathematical model in Section II, two linearity indices are proposed as measures to evaluate the degree of linearity of 2-D position measurement in Section III. The measures are found to be effective when comparing the linearities of the 2-D tetralateral PSD and a modified tetralateral PSD, which is a square PSD with L-shape electrodes [9]. In Section IV, we present a systematic procedure for designing new estimation formulas with simple combinations of electrode currents, and the coefficients can be optimized to meet any specified linearity characterized by the linearity indices. Following the procedure, a simple estimation formula is proposed for the 2-D tetralateral PSD with very significant linearity improvement. More importantly, the design procedure and the derived formula can be applied directly to linearity improvement of other 2-D PSDs like position-sensitive silicon detector (PSSD) for heavy-ion measurements [16], [17], where only the coefficients in the formula need to be recustomized before application. In Section V, through simulation, it has been found that the position measurement error is less than 1% in 32% of the effective PSD area based on the new formula as compared to 6% of the effective area based on the conventional formula. Quantitative comparison indicates that the performance of the new formula is better than any other modification of the 2-D tetralateral PSD while maintaining a simple PSD structure. To test the performance in real implementations, an experiment has been conducted, and the result is very impressive. Finally, conclusions are drawn in Section VI.

II. Mathematical Model

The basic structure of the 2-D tetralateral PSD is shown in Fig. 1(a), and the origin of the coordinate system is the center of the sensor. The cross section of the PSD and its physical principle are shown in Fig. 1(b). Lucovsky [18] has shown that the electric potential distribution in steady state is governed by the following equation when it is fully reverse biased:

$$\nabla^2 U(x,y) = -\frac{\rho_d}{w_d} I_s(x,y) \tag{1}$$

where $U(x,y)$ is the electric potential; ∇^2 denotes the Laplace operator; $I_s(x,y)$ is the generated photocurrent, which is proportional to the light illumination; ρ_d is the resistivity; and w_d is the thickness of the junction. In many applications, the spot size is usually small after focusing, and the incident light on the sensor can be regarded as a point. Then, the generated photocurrent can be expressed as a Dirac function:

$$I_s(x,y) = I_0 \delta(x_0 - x)\delta(y_0 - y) \tag{2}$$

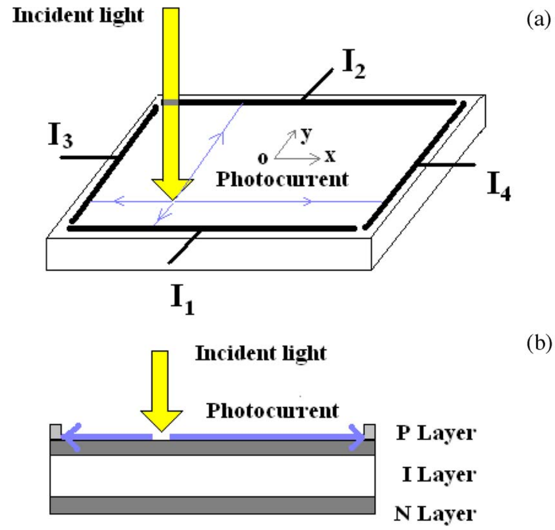


Fig. 1. (a) Basic structure of the PSD. (b) Cross section of the PSD and its physical principle.

where (x_0, y_0) denotes the position of the incident light on the PSD. The length of the electrode is assumed to be one unit in this paper for simplicity. In that case, the electric potential can be expressed explicitly when the boundary condition for the 2-D tetralateral PSD applies [4]:

$$U(x,y) = I_0 \frac{\rho_d}{\omega_d} \times \sum_{m=-\infty}^{\infty} \sum_{n=-\infty}^{\infty} \frac{\sin(mx\pi)\sin(ny\pi)\sin(mx_0\pi)\sin(ny_0\pi)}{(n^2+m^2)\pi^2}. \tag{3}$$

The electrical currents collected at the electrodes can be calculated explicitly from the following integrals, where p_i is the normal to each electrode and l_1–l_4 refer to the lengths of the four electrodes [4]:

$$I_i = \frac{\omega_d}{\rho_d} \int_{l_i} \left.\frac{\partial U}{\partial p}\right|_{p=p_i} dl_i, \qquad i = 1, 2, 3, 4. \tag{4}$$

The conventional position estimation formula can be found in the literature [4]. The position estimate $\hat{X}$ in the x-direction and $\hat{Y}$ in the y-direction will be dependent on the position of the incident light on the sensor, and they are given by:

$$\hat{X}(x_0,y_0) = k\frac{I_4 - I_3}{I_4 + I_3}$$
$$\hat{Y}(x_0,y_0) = k\frac{I_2 - I_1}{I_2 + I_1} \tag{5}$$

where k is the linearity constant in the formula. There is no estimation error when the spot is on the PSD center:

$$\hat{X}(0,0) = 0$$
$$\hat{Y}(0,0) = 0. \tag{6}$$

III. Linearity Indices

It is found that when the light spot is near the PSD center, $\hat{X}$ is proportional to x_0 and that $\hat{Y}$ is proportional to y_0 approximately. This approximation becomes inaccurate when x_0 and y_0 become larger, i.e. when the spot is further away from the center. These observations can be well explained by Taylor series expansion in two dimensions, e. g. $\hat{X}$ is expanded as:

$$\begin{aligned}\hat{X}(x_0, y_0) &= \left[x_0 \left.\frac{\partial \hat{X}}{\partial x_0}\right|_{x_0=0, y_0=0} + y_0 \left.\frac{\partial \hat{X}}{\partial y_0}\right|_{x_0=0, y_0=0} \right] \\ &+ \frac{1}{2!}\left[x_0^2 \left.\frac{\partial^2 \hat{X}}{\partial x_0^2}\right|_{x_0=0, y_0=0} + 2x_0 y_0 \left.\frac{\partial^2 \hat{X}}{\partial x_0 \partial y_0}\right|_{x_0=0, y_0=0} \right. \\ &\left. + y_0^2 \left.\frac{\partial^2 \hat{X}}{\partial y_0^2}\right|_{x_0=0, y_0=0} \right] + \cdots. \end{aligned} \tag{7}$$

When the light spot is near the PSD center, x_0 and y_0 are small, and it can be shown that $(\partial \hat{X}/\partial y_0)|_{x_0=0,y_0=0} = 0$, so the first term in the expression dominates, and that is why, it is observed that $\hat{X}$ is linear with respect to x_0 approximately. However, contributions from the nonlinear terms increase when x_0 and y_0 are larger and $\hat{X}$ is no longer linear with respect to x_0. This will give measurement errors, and one is unable to know x_0 accurately from the information of $\hat{X}$ because of the implicit relationship. The estimate will be more accurate only if the degree of linearity is enhanced. In addition, the lower order nonlinear terms in the expression, as compared to higher order terms that are neglected, usually dominate the contributions to the nonlinearity. Thus, the nonlinearity can be reduced if the lower order terms have zero or small values. In this sense, these lower order terms can be used to define linearity indices to measure the degree of linearity for different estimation schemes of the 2-D PSD. The linearity indices also serve the purpose of measuring the estimation errors. Before we introduce the linearity indices, it must be noted that the estimation formulas for all types of 2-D PSD have the property of center symmetry. This is because every 2-D PSD has a center-symmetric structure, which means that estimation inaccuracy is center symmetric too. Thus, only the analysis of position estimate $\hat{X}$ will be given in detail in this paper because the same analysis also applies to $\hat{Y}$. The property of center symmetry is expressed as:

$$\begin{aligned}\hat{X}(x_0, y_0) &= \hat{X}(x_0, -y_0) \\ \hat{X}(x_0, y_0) &= -\hat{X}(-x_0, y_0)\end{aligned} \tag{8}$$

or equivalently:

$$\begin{aligned}\left.\frac{\partial^{2m+n+1}\hat{X}}{(\partial y_0)^{2m+1}(\partial x_0)^n}\right|_{y_0=0} &= 0, \qquad m, n = 0, 1, 2, \ldots \\ \left.\frac{\partial^{2m+n}\hat{X}}{(\partial x_0)^{2m}(\partial y_0)^n}\right|_{x_0=0} &= 0, \qquad m, n = 0, 1, 2, \ldots.\end{aligned} \tag{9}$$

We propose the following two linearity indices for $\hat{X}$ as measures of linearity in the x- and y-directions:

$$LH_x^n \triangleq \left| \frac{\left.\dfrac{\partial^n \hat{X}}{\partial x_0^n}\right|_{x_0=0, y_0=0}}{\left.\dfrac{\partial \hat{X}}{\partial x_0}\right|_{x_0=0, y_0=0}} \right| \tag{10}$$

$$LV_x^n \triangleq \left| \frac{\left.\dfrac{\partial^n \hat{X}}{\partial y_0^{n-1}\partial x_0}\right|_{x_0=0, y_0=0}}{\left.\dfrac{\partial \hat{X}}{\partial x_0}\right|_{x_0=0, y_0=0}} \right| \tag{11}$$

where n is an integer such that all $LH_x^i = 0$ for $2 \leq i \leq n-1$ and $LH_x^n \neq 0$. It should be noted that $LH_x^2 = 0$ is always satisfied when it is center symmetric. The position measurement $\hat{X}$ satisfying this property is referred to as having nth-order linearity with respect to the x-direction. For example, it can be calculated that $LH_x^2 = 0$ while $LH_x^3 \neq 0$ for the 2-D tetralateral PSD with the conventional formula, and so, the position measurement $\hat{X}$ is referred to as having third-order linearity with respect to the x- direction. Similarly, the nth order of linearity with respect to the y direction is defined as LV_x^n, where n is an integer such that all $LV_x^i = 0$ for $2 \leq i \leq n-1$ and $LV_x^n \neq 0$. Higher order linearity indices indicate better linearity since more lower order terms have zero values. When linearity indices have the same order, linearity indices with smaller values would indicate better linearity. In the same way, linearity indices for the position measurement $\hat{Y}$ can be defined as well. The linearity indices for the 2-D tetralateral PSD with the conventional formula can be calculated as:

$$LH_x^3 = \left| \frac{\left.\dfrac{\partial^3 \hat{X}(x_0, y_0)}{\partial x_0^3}\right|_{x_0=0, y_0=0}}{\left.\dfrac{\partial \hat{X}(x_0, y_0)}{\partial x_0}\right|_{x_0=0, y_0=0}} \right| = LV_y^3 = \left| \frac{\left.\dfrac{\partial^3 \hat{Y}(x_0, y_0)}{\partial y_0^3}\right|_{x_0=0, y_0=0}}{\left.\dfrac{\partial \hat{Y}(x_0, y_0)}{\partial y_0}\right|_{x_0=0, y_0=0}} \right| = 19.5 \tag{12}$$

$$LV_x^3 = \left| \frac{\left.\dfrac{\partial^3 \hat{X}(x_0, y_0)}{\partial y_0^2 \partial x_0}\right|_{x_0=0, y_0=0}}{\left.\dfrac{\partial \hat{X}(x_0, y_0)}{\partial x_0}\right|_{x_0=0, y_0=0}} \right| = LH_y^3 = \left| \frac{\left.\dfrac{\partial^3 \hat{Y}(x_0, y_0)}{\partial x_0^2 \partial y_0}\right|_{x_0=0, y_0=0}}{\left.\dfrac{\partial \hat{Y}(x_0, y_0)}{\partial y_0}\right|_{x_0=0, y_0=0}} \right| = 1.9. \tag{13}$$

Since the linearity indices defined in (10) and (11) are universal for all 2-D PSD measurements, the linearity indices for the modified tetralateral PSD with L-shape electrodes can also be calculated, and they are:

$$LH_x^3 = \left| \frac{\left. \dfrac{\partial^3 \hat{X}(x_0, y_0)}{\partial x_0^3} \right|_{x_0=0, y_0=0}}{\left. \dfrac{\partial \hat{X}(x_0, y_0)}{\partial x_0} \right|_{x_0=0, y_0=0}} \right| = LV_y^3 = \left| \frac{\left. \dfrac{\partial^3 \hat{Y}(x_0, y_0)}{\partial y_0^3} \right|_{x_0=0, y_0=0}}{\left. \dfrac{\partial \hat{Y}(x_0, y_0)}{\partial y_0} \right|_{x_0=0, y_0=0}} \right| = 4.3 \quad (14)$$

$$LV_x^3 = \left| \frac{\left. \dfrac{\partial^3 \hat{X}(x_0, y_0)}{\partial y_0^2 \partial x_0} \right|_{x_0=0, y_0=0}}{\left. \dfrac{\partial \hat{X}(x_0, y_0)}{\partial x_0} \right|_{x_0=0, y_0=0}} \right| = LH_y^3 = \left| \frac{\left. \dfrac{\partial^3 \hat{Y}(x_0, y_0)}{\partial x_0^2 \partial y_0} \right|_{x_0=0, y_0=0}}{\left. \dfrac{\partial \hat{Y}(x_0, y_0)}{\partial y_0} \right|_{x_0=0, y_0=0}} \right| = 6.7. \quad (15)$$

The quantitative calculation shows that the position estimate $\hat{X}$ for the 2-D tetralateral PSD has a poorer linearity than its modified version in the x-direction but better in the y-direction. This conclusion is supported by the simulation result. In the simulation, the light spot is moved from the center in steps of 0.05 in both directions inside the central square of 0.8×0.8. Then, position estimates are shown for both 2-D tetralateral PSD and its modified counterpart in Figs. 2 and 3, respectively. If the estimation is perfect without inaccuracy, they will form a regular grid pattern. In the simulation, when the light spot moves in steps along the x-direction from the center, the distance between adjacent position estimates will be nearly equal if $\hat{X}$ has small nonlinearity in the x-direction. From Figs. 2 and 3, it can be seen that the distance between adjacent position estimates for the 2-D tetralateral PSD varies more significantly than that for the modified 2-D tetralateral PSD, which means a poorer linearity for $\hat{X}$ in the x-direction. When the light spot moves in steps along the y-direction for a fixed x-coordinate from top to bottom, the position estimates should be well aligned vertically if $\hat{X}$ has small nonlinearity in the y-direction. From Figs. 2 and 3, it can be seen that the alignment is poorer in the modified 2-D tetralateral PSD, which verifies that its position estimate $\hat{X}$ has a larger nonlinearity in the y-direction. These phenomena are well explained by our linearity indices, and hence, our proposal of these linearity indices as measures of the linearity of the 2-D PSD is justified.

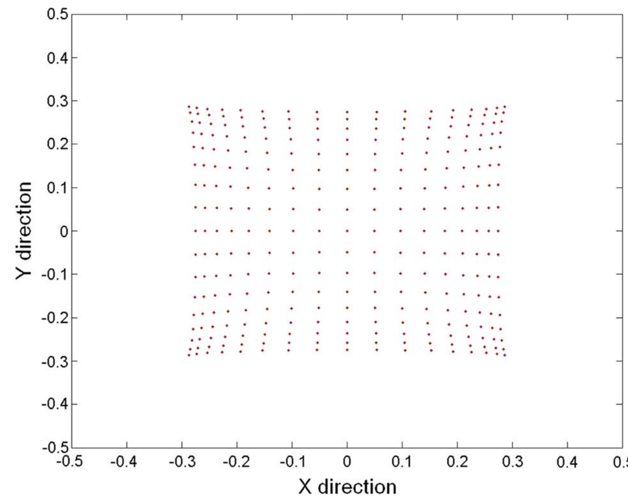


Fig. 2. Simulation result of the 2-D tetralateral PSD using the conventional formula.

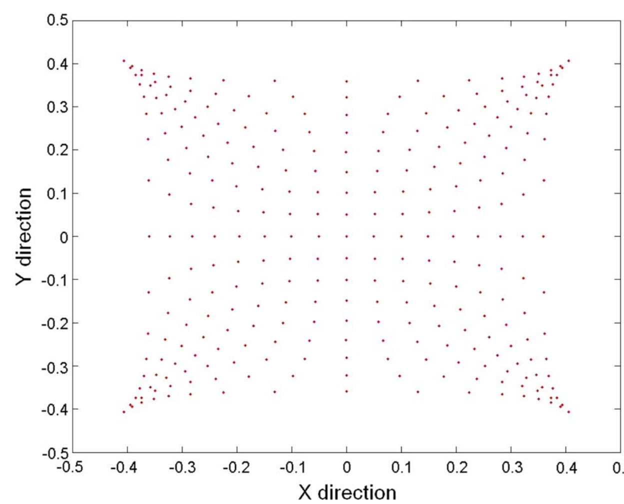


Fig. 3. Simulation result of the modified 2-D tetralateral PSD.

IV. New Estimation Formula

The difficulty of achieving an accurate estimation lies in the implicit nonlinear relationship between the position estimate and the real incident position. It is impossible to recover exactly the real incident position based on the information of the four current outputs from the electrodes, regardless of any tetralateral electrodes and boundary design. The only possibility is to improve the estimation accuracy by improving the degree of linearity. While creative designs of the electrodes and boundary [9] have improved the linearity, further improvement is rather difficult, and the technique cannot be applied to other 2-D PSDs like PSSD for different applications. In this section, through the derivation of a new formula for the 2-D tetralateral PSD, a systematic procedure is presented to design estimation formulas to meet any degree of linearity for the 2-D PSD. Due to the property of center symmetry, only the derivation for $\hat{X}$ is given, and that for $\hat{Y}$ can be obtained in the same manner. The desirable estimation formula, which is based on the four output currents, should have simple arithmetic for easy implementation. We first examine a formula that involves the ratio of two linear combinations of the four currents:

$$\hat{X}(x_0, y_0) = \frac{\sum_{i=1}^{4} a_i I_i}{\sum_{i=1}^{4} b_i I_i}. \tag{16}$$

The coefficients are to be optimized for linearity improvement. First of all, the property of center symmetry must be satisfied, $\partial \hat{X}/\partial x_0$ must be nonzero to ensure linearity, and $\hat{X}$ should not be infinity when the light spot is on the PSD. To fulfill all these preconditions, we have:

$$\hat{X}(x_0, y_0) = \frac{a_4(I_4 - I_3)}{b_1(I_1 + I_2) + b_4(I_3 + I_4)}.$$

There is still flexibility for further linearity improvement. Guided by the proposed linearity indices, since $\partial^3 \hat{X}(x_0, y_0)/\partial x_0^3|_{x_0=0,y_0=0} \neq 0$ and $\partial^4 \hat{X}(x_0, y_0)/\partial x_0^4|_{x_0=0,y_0=0} = 0$ in general, we can achieve fifth order of linearity with respect to the x-direction by choosing coefficients to satisfy:

$$\frac{\partial^3 \hat{X}(x_0, y_0)}{\partial x_0^3}|_{x_0=0,y_0=0} = 0 \tag{17}$$

which requires $b_4 = 1.72\ b_1$. At this stage, there is no more flexibility for further linearity improvement because the coefficients a_4 and b_1 only relate to the linearity slope and not the nonlinearity. With the aforementioned $\hat{X}(x_0, y_0)$, the linearity index for $\hat{X}$ with respect to the y-direction can be calculated as $|((\partial^3 \hat{X}(x_0,y_0)/\partial x_0 \partial y_0^2)|_{x_0=0,y_0=0})/(\partial \hat{X}(x_0,y_0)/\partial x_0)|_{x_0=0,y_0=0})| = 5.5$, which is poorer than the linearity index calculated in (13) for $\hat{X}$ with respect to the y-direction when the conventional formula is used. Thus, a new estimation formula with more flexibility terms is needed for linearity improvement in $\hat{X}$ with respect to both directions.

In general, any degree of linearity improvement for the 2-D PSD is achievable by optimizing the coefficients of a formula that has enough flexibility terms to set to zero the corresponding nonlinearity terms. However, this will give rise to a complex formula that is difficult to implement. A tradeoff between linearity improvement and complexity of the formula is required while satisfying the preconditions. Our aim is to achieve fifth-order linearity with respect to both x- and y-directions for position measurement $\hat{X}$. For this purpose, we made use of the following formula:

$$\hat{X}(x_0, y_0) = \frac{\sum_{i=1}^{4} a_i I_i}{\sum_{i=1}^{4} b_i I_i} \frac{\sum_{i=1}^{4} c_i I_i}{\sum_{i=1}^{4} d_i I_i}. \tag{18}$$

With optimized coefficients, the following estimation formulas have been found to satisfy the requirement:

$$\hat{X}(x_0, y_0) = \frac{k(I_4 - I_3)}{I_{\text{sum}} - 1.02(I_2 - I_1)} \times \frac{0.7(I_1 + I_2) + I_{\text{sum}}}{I_{\text{sum}} + 1.02(I_2 - I_1)}$$
$$\hat{Y}(x_0, y_0) = \frac{k(I_2 - I_1)}{I_{\text{sum}} - 1.02(I_4 - I_3)} \times \frac{0.7(I_3 + I_4) + I_{\text{sum}}}{I_{\text{sum}} + 1.02(I_4 - I_3)} \tag{19}$$

where $I_{\text{sum}} = \sum_{i=1}^{4} I_i$.

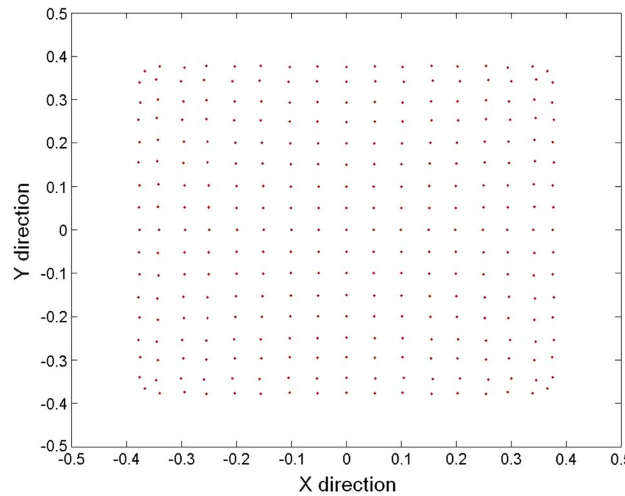


Fig. 4. Estimation result of the 2-D tetralateral PSD using the proposed new formula.

TABLE I
DISTORTION COMPARISON AMONG DIFFERENT LINEARITY IMPROVEMENT SCHEMES

	Percentage of area with distortion less than			
	1.0%	2.0%	3.0%	4.0%
Tetra-lateral	6.0	10.4	13.1	16.2
Pin-cushion	6.9	13.5	20.8	31.6
Clover	17.1	52.7	67.6	76.4
Tetra-lateral (new formula)	31.5	56.4	71.6	84.1

V. SIMULATION AND EXPERIMENTAL RESULTS

To verify the performance of the new formulas, simulations and one experiment are conducted. In the simulation, the light spot is assumed to move in steps, following the same procedure as that in the previous simulation. The simulation result is shown in Fig. 4, and the estimation accuracy is almost perfect. For a quantitative comparison, the percentages of the effective area with measurement errors less than 1%, 2%, 3%, and 4% are tabulated in Table I. For a fair comparison, the effective area defined here is the same as Wang's definition [9], bounded by a central square with a side length of 10/12. Measurement errors are defined in [19]: $D = x_0 - \hat{X}(x_0, y_0)/x_0$, where $\hat{X}(x_0, y_0)$ is the normalized estimate in the x-direction and x_0 is the real position in the x-direction. Comparisons are summarized in the table, where some data are from Wang's calculation [9]. It can be seen that the percentage of the area with less than 1% measurement error, considered as error-free area, has been greatly enlarged to 32% as compared to 6% based on the conventional formula. This is quite significant, and it even has a better performance than the clover PSD.

In addition to simulations, an experiment was performed to test the new formula in the real environment. The experimental setup consists of a 2-D tetralateral PSD (S1200 from Hamamatsu Photonics), a He–Ne laser tube with adjustable intensity, and a motorized X–Y–Z stage. The active area of the PSD is 13 mm × 13 mm. The motorized stage has a motion accuracy of 50 nm. A special lens system is designed to reduce the spot diameter to 100 μm in front of the laser. The digital

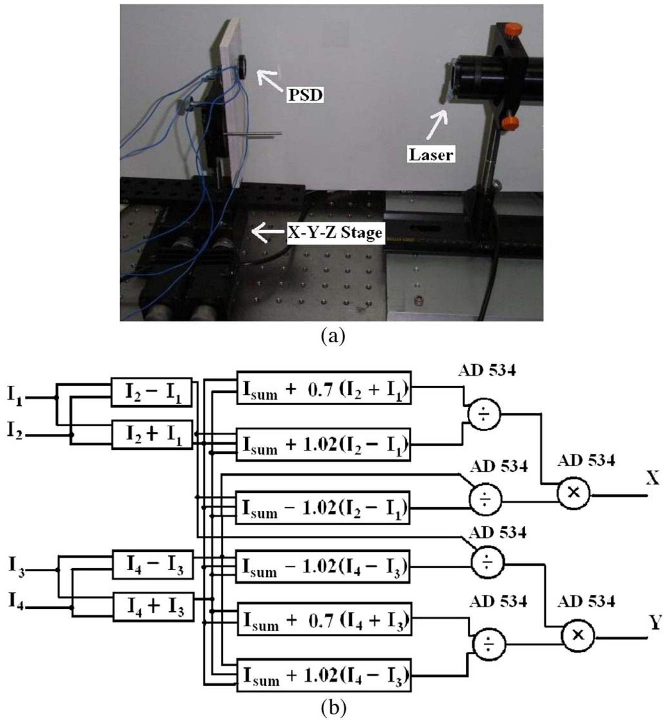


Fig. 5. (a) Digital image of the experimental setup. (b) Schematic of the electrical circuit to implement our new formula.

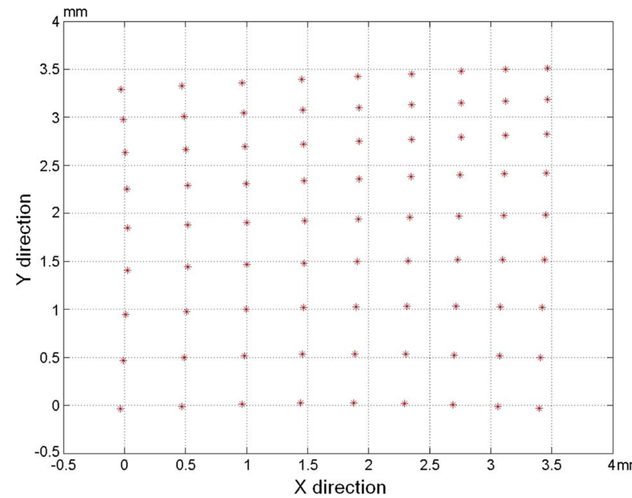


Fig. 6. Experimental result for position measurements by using the 2-D tetralateral PSD with the conventional formula.

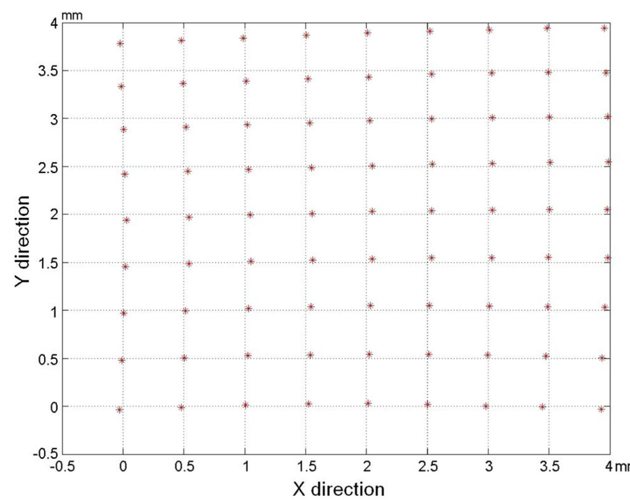


Fig. 7. Experimental result for position measurements by using the 2-D tetralateral PSD with the new formula.

image of the experimental setup is shown in Fig. 5(a). Before the experiment, the PSD is fixed on a target plane holder, which is well aligned with a tilt alignment system. The laser becomes stable after it is turned on for a while. The whole experiment is carried out in a dark environment. During the experiment, the center of the PSD is first sought by locating the zero value in the position readings. Then, electrode currents are recorded when the laser spot is scanning the PSD line by line with a step size of 0.5 mm in both directions. The conventional formula and our new formula are both implemented in electrical circuits with analog devices like operational amplifier for addition and subtraction and analog computational units (AD534) for multiplication and division. The schematic of the electrical circuit to implement our new formula is shown in Fig. 5(b). The electrical circuit to implement our new formula has six AD534 units, which is slightly more complex than the four AD534 units deployed in the conventional formula. Thus, the complexity and the additional cost required to achieve linearity improvement are slightly increased. For simplicity, only one quadrant of the PSD is tested as it is center symmetric. Estimates processed via the conventional and new formulas are shown in Figs. 6 and 7, respectively. The result shows that a good linearity is achieved in the real environment by using our new formula.

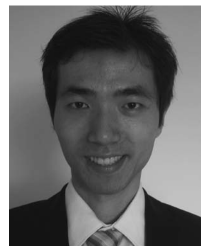
VI. Conclusion

In this paper, we have introduced two linearity indices as measures of linearity or, equivalently, estimation accuracy for 2-D PSDs. They are very useful for explaining the difference in estimation accuracy of the 2-D tetralateral PSD and its modified version. More importantly, a systematic procedure has been proposed to design new estimation formulas for the 2-D tetralateral PSD to achieve any degree of linearity characterized by the linearity indices. Following the procedure, a new formula with a simple structure has been derived to achieve fifth-order linearity with respect to both x- and y-directions for $\hat{X}$ and $\hat{Y}$. An experiment was performed to illustrate very good linearity improvement in the real environment, making it much more superior than any other competitors. Nowadays, there is a strong demand for 2-D PSDs with low cost, low noise level, and fast response to provide accurate measurement for both research and industry applications. With our new formula, the 2-D tetralateral PSD will offer an effective solution to meet these requirements. More significantly, the design procedure can be applied directly to linearity improvement of other 2-D PSDs, where only the coefficients need to be recustomized.

References

[1] J. T. Wallmark, "A new semiconductor photocell using lateral photo-effect," *Proc. IRE*, vol. 45, no. 4, pp. 474–483, Apr. 1957.
[2] W. Schotty, "Ueber den Entsstehungsort der Photoelektronen in Kupfer-Kupferoxydulphotozellen," *Phys. Z.*, vol. 31, pp. 913–925, Nov. 1930.

[3] W. P. Connors, "Lateral photodetector operating in the fully reverse-biased mode," *IEEE Trans. Electron Devices*, vol. ED-18, no. 8, pp. 591–596, Aug. 1971.
[4] H. J. Woltring, "Single- and dual-axis lateral photodetectors of rectangular shape," *IEEE Trans. Electron Devices*, vol. ED-22, no. 8, pp. 581–590, Aug. 1975.
[5] A. Mäkynen, J. T. Kostamovaara, and R. A. Myllylä, "Displacement sensing resolution of position-sensitive detectors in atmospheric turbulence using retroreflected beam," *IEEE Trans. Instrum. Meas.*, vol. 46, no. 5, pp. 1133–1136, Oct. 1997.
[6] Y. Y. Cha and D. G. Gweon, "A calibration and range-data extraction algorithm for an omnidirectional laser range finder of a free-ranging mobile robots," *Mechatronics*, vol. 6, no. 6, pp. 665–689, Sep. 1996.
[7] H. Hoff and J. Bargon, "A vibrating optical fiber tip as a new sensor to monitor organic vapors," *Sens. Actuators B, Chem.*, vol. 67, no. 1/2, pp. 29–35, Aug. 2000.
[8] G. Meyer and N. M. Amer, "Novel optical approach to atomic force microscopy," *Appl. Phys. Lett.*, vol. 53, no. 12, pp. 1045–1047, Sep. 1988.
[9] W. Wang and I. J. Busch-Vishniac, "The linearity and sensitivity of lateral effect position sensitive devices—An improved geometry," *IEEE Trans. Electron Devices*, vol. 36, no. 11, pp. 2475–2480, Nov. 1989.
[10] E. Silva and K. Vliet, "Robust approach to maximize the range and accuracy of force application in atomic force microscopes with nonlinear position-sensitive detectors," *Nanotechnology*, vol. 17, no. 21, pp. 5525–5530, Nov. 2006.
[11] *Position-Sensitive Detectors*, Hamamatsu Photonics K.K., Solid State Div., Hamamatsu City, Japan, Product Catalog, 1987.
[12] W. Wang, "Clover design lateral effect position-sensitive device," U.S. Patent 4 887 140, Dec. 12, 1989.
[13] J. W. Noorlag, "Quantitative analysis of effects causing nonlinear position response in position-sensitive photodetectors," *IEEE Trans. Electron Devices*, vol. ED-29, no. 1, pp. 158–161, Jan. 1982.
[14] X. Wang and M. Ye, "Modeling and non-linear correction of two-dimension photoelectric position sensitive detector," *Proc. SPIE*, vol. 4919, pp. 452–460, Oct. 2002.
[15] M. Huang, L.-Z. Shi, Y.-X. Wang, Y. Ni, Z.-Q. Li, and H.-F. Ding, "Development of a new signal processor for tetralateral position sensitive detector based on single-chip microcomputer," *Rev. Sci. Instrum.*, vol. 77, no. 8, p. 083 301, Aug. 2006.
[16] M. Bruno, M. D'Agostino, M. L. Fiandri, E. Fuschini, P. M. Milazzo, S. Ostuni, F. Gramegna, I. Iori, L. Manduci, A. Moroni, R. Scardaoni, P. Buttazzo, G. V. Margagliotti, and G. Vannini, "Position determination and resolution of two-dimensional position-sensitive solid-state detectors," *Nucl. Instrum. Methods Phys. Res. A, Accel. Spectrom. Detect. Assoc. Equip.*, vol. 311, no. 1/2, pp. 189–194, Jan. 1992.
[17] R. L. Cowin and D. L. Watson, "Surface charge flow in two-dimensional position sensitive silicon detectors," *Nucl. Instrum. Methods Phys. Res. A, Accel. Spectrom. Detect. Assoc. Equip.*, vol. 399, no. 2/3, pp. 365–381, Nov. 1997.
[18] G. Lucovsky, "Photoeffects in nonuniformly irradiated p-n junctions," *J. Appl. Phys.*, vol. 31, no. 6, pp. 1088–1095, Jun. 1960.
[19] G. Petersson and L. Linholm, "Position sensitive light detectors with high linearity," *IEEE J. Solid-State Circuits*, vol. SSC-13, no. 3, pp. 392–399, Jun. 1978.



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